Level: 4000

For your convenience, my code is also available on GitHub:
https://github.com/Youssef-Chip/data-analytics/blob/main/final_project/final_project.ipynb

This Jupyter Notebook contains all of my R code, Markdown documentation, and plots. However, I noticed GitHub struggles to render R cells perfectly, so I recommend reviewing the submitted file locally for the best experience.

1. Abstract and Introduction

For this project, I attempted to build a machine learning pipeline capable of predicting the outcomes of international football matches. I watch many football matches, and since the biggest sporting event in the world is coming to North America this summer, I wanted my pipeline to simulate and predict the winner of the 2026 FIFA World Cup.

My initial hypothesis was that a standard win/loss classification model is insufficient for simulating a tournament like the World Cup because it fails to capture goal differentials, which are crucial for tie-breakers. Therefore, I hypothesized that predicting Expected Goals, using a regression model and converting those into exact scoreline probabilities using a Poisson distribution would yield a more realistic and highly accurate simulation.

A good outcome for this project isn't necessarily a model that guesses everything with 100% certainty, but one that beats the baseline strategy of always picking the higher-ranked team. This way, when the tournament starts, I can track my bracket and see exactly how close my simulations were to reality.

2. Data Description and preliminary analysis

The data used in this project is from the "International football results from 1872 to the Present" dataset, hosted on Kaggle and maintained by Mart Jürisoo. I chose this dataset (Jürisoo, 2026) because it is well documented, regularly updated, and contains over 49,000 historical men's international football results.

The primary dataset, **results.csv**, contains the following key variables:

- **Temporal Data:** The date of the match
- **Categorical Data:** home_team, away_team, neutral, tournament, city, and country. There are 191 unique tournament types recorded (from minor friendlies to the FIFA World Cup). The neutral column indicates whether the match was played on neutral territory, which is essential for quantifying home-field advantage.
- **Continuous Data:** home_score and away_score, representing the goals scored by each team.

In addition to the main results table, the author provided supporting datasets that I incorporated during my preliminary analysis and cleaning phase:

Former Names Mapping: I used the **former_names.csv** dataset to address teams that have changed their names over time, to ensure a country's historical timeline remains consistent.

Shootout Resolution: In knockout tournaments, drawn matches end in penalty shootouts. The initial dataset only tracks the score at the end of extra time. I merged the **shootouts.csv** dataset to show the true winners of these drawn matches.

For the preliminary analysis and data formatting phase, I first converted the date column from strings to standard Date objects to facilitate chronological operations. I also added a year column for later aggregations during the exploratory data analysis phase. Then, I converted the neutral column from boolean values to numeric values (1 and 0). Finally, I checked for missing values across the merged dataframes: as expected, the only null values generated were in the newly merged shootout_winner column for matches that ended in regular time and did not require a penalty shootout. Otherwise, the core dataset was clean and therefore required no imputation for missing scores or team names before moving into the exploratory visualization phase.

- **Figure 1: 2022 World Cup Final record showing the true match winner after merging shootout data.**

A data.frame: 1 x 11

	date	home_team	away_team	home_score	away_score	tournament	city	country	neutral	year	shootout_winner
	<date>	<chr>	<chr>	<int>	<int>	<chr>	<chr>	<chr>	<dbl>	<dbl>	<chr>
45750	2022-12-18	Argentina	France	3	3	FIFA World Cup	Lusail	Qatar	1	2022	Argentina

- **Figure 2: Summary of missing values across the cleaned dataset**

```
date:          0 home_team:          0 away_team:          0 home_score:          0 away_score:          0 tournament:          0
```

```
city:          0 country:          0 neutral:          0 year:          0 shootout_winner:          48407
```

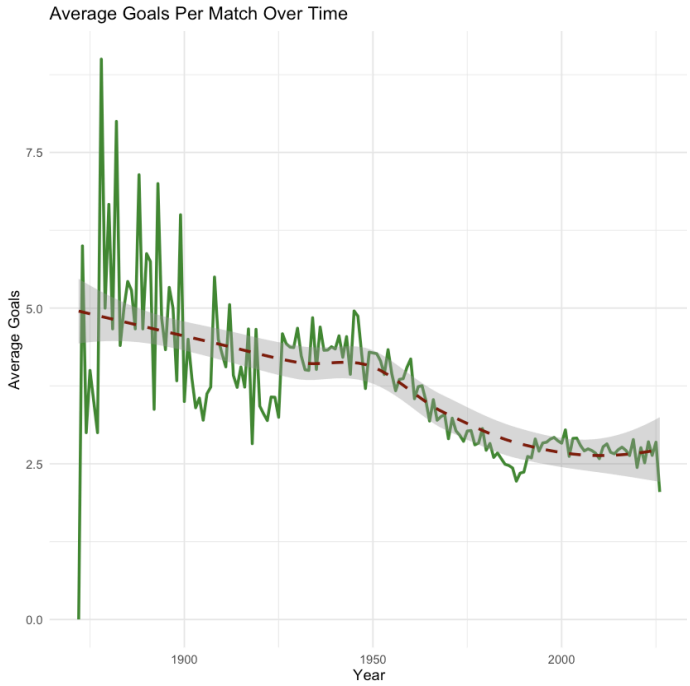
3. Exploratory Analysis

Before building advanced features, I addressed a character encoding issue. I noticed that "Copa América" was not displaying or accessed correctly due to the accented 'é'. I resolved this by using its raw character code (\303\251). With the text standardized, I created a new binary column called is_major_tournament to flag the most important competitions: the FIFA World Cup, UEFA Euro, Copa América, African Cup of Nations, and AFC Asian Cup.

Next, I focused on feature engineering to capture team strength and recent form: First, I implemented a continuous historical rating system using the elo package to give the model a baseline understanding of team strength. Second, I calculated 5-game rolling metrics to capture each team's recent form. I added six new pre-match features to every row: last_5_points, last_5_gs (goals scored), and last_5_gc (goals conceded) for both the home and away teams.

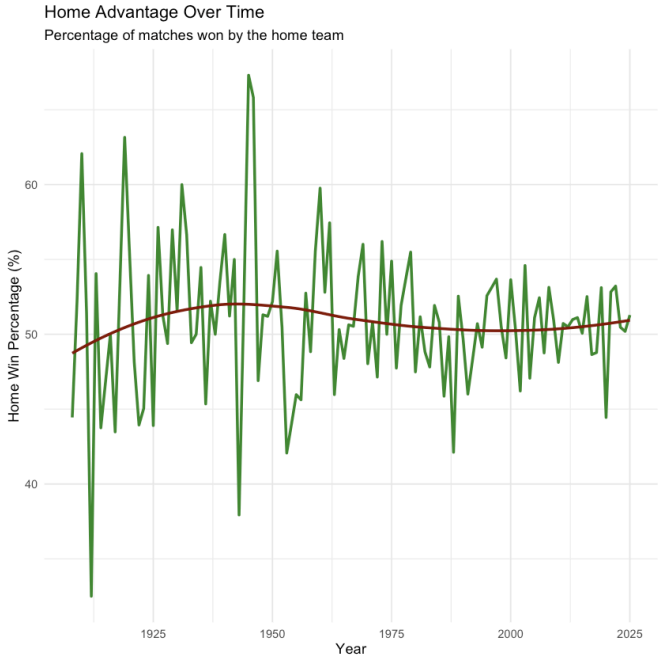
Then for the exploratory data analysis, I analyzed several distributions:

- **Figure 3: Average Goals Over Time**



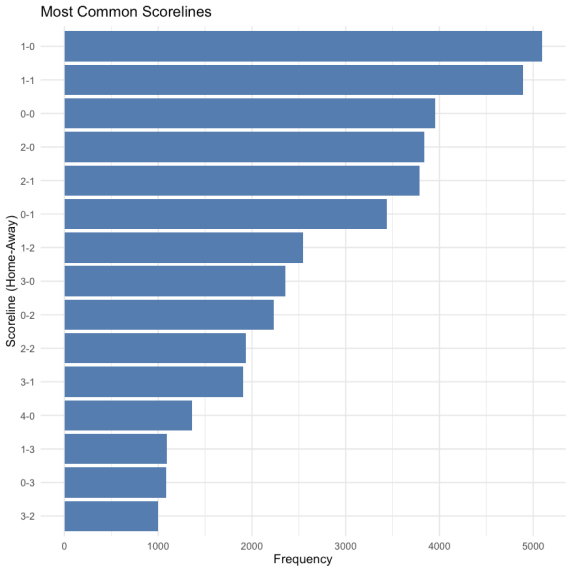
High scoring in early football was too inconsistent to be reliable. To ensure the model doesn't over-predict goals based on the 1800s, I limited the final training data strictly to matches from 1998 onward, when scoring rates stabilized.

- **Figure 4: Home Field Advantage**



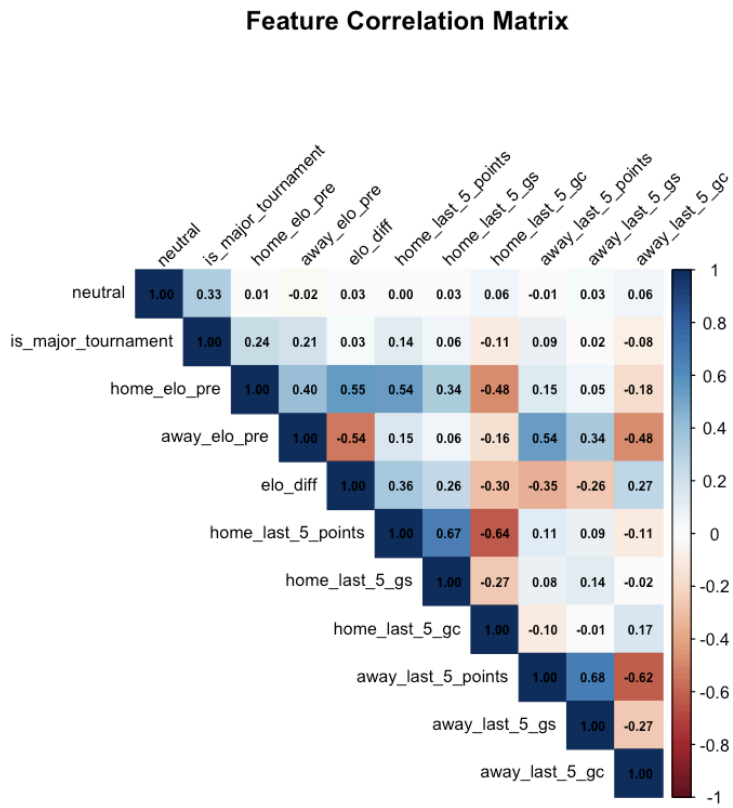
This graph proves that home advantage is real. Because the home team wins roughly 50% of the time in a sport with three possible outcomes, they hold a distinct edge over the away team.

- **Figure 5: Scoreline Frequencies**



The most common scorelines are incredibly low, proving that goal data is heavily right-skewed. A standard linear regression would struggle with this because it's not a normal distribution, which mathematically justified my choice to use a Poisson distribution for the target variables.

- **Figure 6: Feature Correlation**



The correlation matrix revealed natural relationships (recent form vs. goals scored), but more importantly, there were no correlations approaching 1.0 or -1.0 among the independent variables. This confirmed that there was no exact duplicate information or multicollinearity in the engineered dataset.

Sources of Bias

One potential source of bias in this dataset is the reliance on historical Elo ratings. A country with a historically dominant team might retain a high Elo rating even if their current golden generation of players has retired which would lead the model to overestimate their current strength. One of the reasons for introducing rolling momentum features was to attempt to mitigate this.

4. Model Development and Application of Models

A standard binary classification model (Win/Loss) is insufficient for simulating a World Cup tournament because it fails to capture goal differences required for tie-breakers. Furthermore, attempting to predict a single "Goal Difference" metric isn't enough because a +1 differential could represent a 1-0 or a 4-3 match.

Therefore, I used XGBoost (Chen & Guestrin, 2016) regressors with a Poisson objective function so that the trees can predict discrete events (goals), where the lowest possible output is 0 (there are no negative goals) and low numbers (0, 1, 2) are highly common, whereas high numbers (5, 6, 7) are extremely rare. While the Poisson objective function strictly follows the mathematical rules of counting whole number events, the final output of the model is a decimal representing the average expected rate of occurrence: Expected Goals. At the end, to predict the scoreline, we translate these into whole number probabilities.

I developed two different models that use different input features to predict Expected Goals:

Model 1: The Baseline Elo Model

- **Features Used:** home_elo_pre, away_elo_pre, elo_diff.
- **Description:** This model relies strictly on historical team strength (Elo) to see if recent momentum actually matters.

Model 2: The Advanced Model

- **Features Used:** home_elo_pre, away_elo_pre, elo_diff, neutral, is_major_tournament, home_last_5_points, home_last_5_gs, home_last_5_gc, away_last_5_points, away_last_5_gs, away_last_5_gc.
- **Description:** This model incorporates venue advantage, tournament importance, team strength, and rolling 5-game momentum statistics to capture a team's current form heading into a match.

The Advanced Model required predicting both Home expected goals and Away expected goals simultaneously to simulate a full match. Therefore, I trained two parallel versions of the model (one for the Home target, one for the Away target) using the exact same advanced feature set. To prevent overfitting, which is a risk with XGBoost, I set a limited tree complexity (max_depth = 4), meaning the model has to make a decision using a maximum of 4 consecutive if/then questions. I also used a subsampling rate of 0.8, which means each decision tree is only allowed to look at a random 80% of the historical matches (which forces the overall model to learn general patterns rather than obsessing over every single data point).

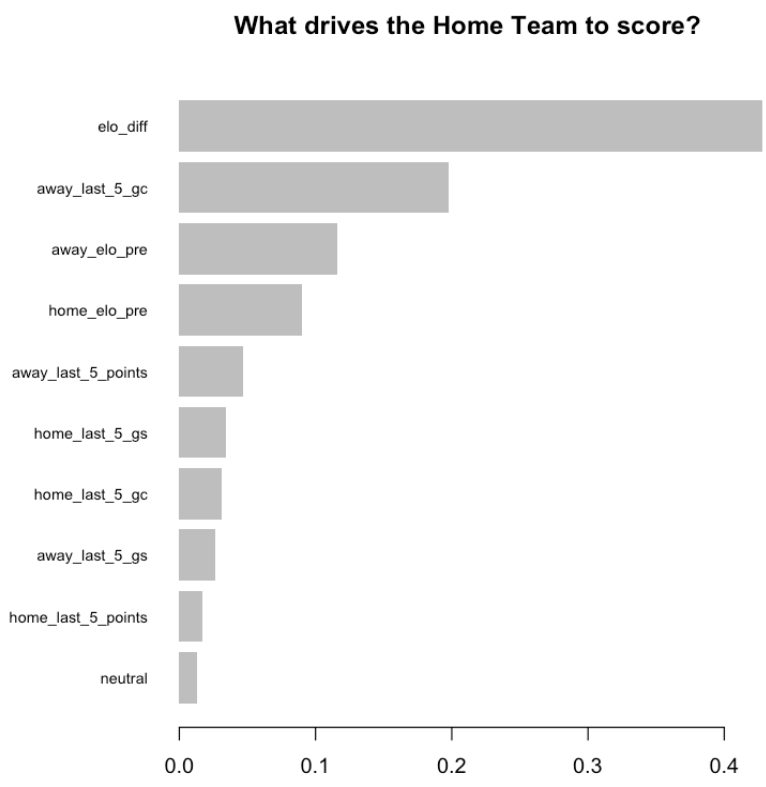
To properly test the hypothesis that recent momentum matters, I trained the Baseline Elo Model using the exact same parameters. However, since it was strictly a baseline comparison and I expected it to perform worse, I only trained a Home Expected Goals regressor so I could compare it directly against the Advanced Home regressor.

Validation and Prevention of Data Leakage: To strictly prevent data leakage and properly validate the models, I used a chronological 3-way split:

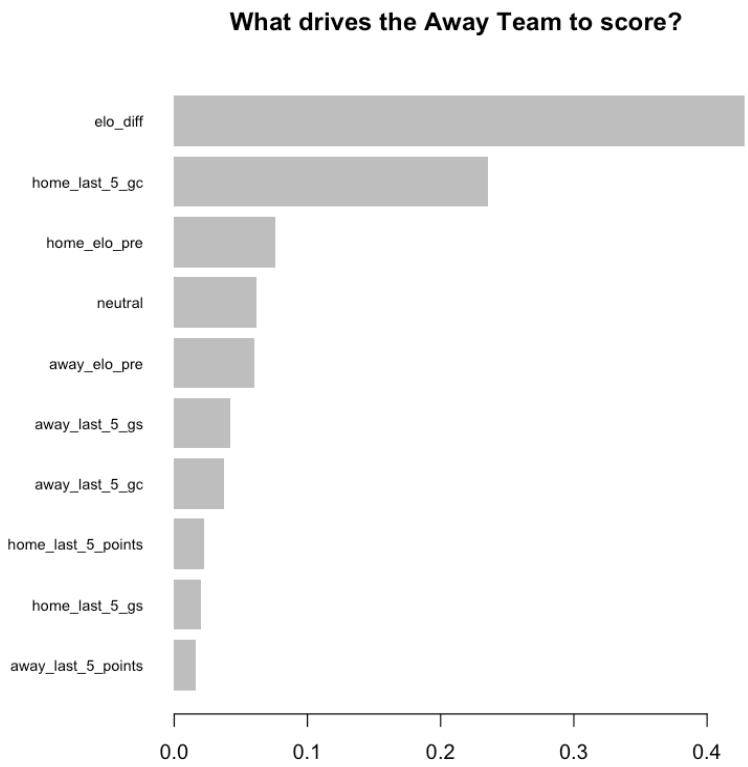
1. **Train:** 1998 through 2017.
2. **Validation:** 2018 through Nov 19, 2022 (used to evaluate the model during training and stop the process early if we see no progress in the loss decreasing after 20 iterations).
3. **Test:** Nov 20, 2022 onwards (From the 2022 FIFA World Cup).

After training the advanced model, I visualized feature importance, which measures how much each feature contributes to the model's prediction accuracy. This revealed the dominance of Elo difference as a feature, as it was by far the most significant feature in predicting goals (accounting for nearly 50% of the information gain for both teams). The second most important feature wasn't a team's offensive momentum, but their opponents defensive vulnerability, which was an interesting insight into modern football.

- **Figure 7: Home Expected Goals Feature Importance**



- **Figure 8: Away Expected Goals Feature Importance**



RMSE and Model Selection

To evaluate the models, I predicted the Expected Goals for the unseen Test set and calculated the Root Mean Squared Error (RMSE). RMSE tells us, on average, how many goals our prediction was off by.

- **Advanced Home expected goals RMSE:** 1.392
- **Advanced Away expected goals RMSE:** 1.137
- **Baseline Home expected goals RMSE:** 1.425

As hypothesized, the Advanced Home Model outperformed the Baseline Home Model. Because the inclusion of recent momentum features successfully reduced the overall prediction error, I proceeded using the Advanced feature set for the final tournament simulations.

Testing the 2022 World Cup

By generating two independent Expected Goal values for every matchup, we can use their Poisson distributions to calculate the precise percentage probability of exact whole-number scorelines. I built a custom Poisson Match Simulator function that takes the Home and Away expected goals and calculates the exact probabilities for scoring 0 through 7 goals.

The function creates a matrix of every possible scoreline combination. By summing specific regions of the matrix (the lower triangle for a Home win, the diagonal for a Draw, and the upper triangle for an Away win), the function returns the exact percentage chance of each match outcome. I applied this simulator to the 2022 World Cup.

For example, here is what the model predicted for the 2022 World Cup Final:

- **Matchup:** Argentina vs France
- **Model Expected Goals:** 1.37 - 1.69
- **Win Probability:** Argentina (31.2%) | Draw (23.7%) | France (45.1%)
- **Actual Score:** 3 - 3 (Draw)

Model Performance

The model achieved **53.1% accuracy** on the unseen 2022 World Cup dataset. In a low-scoring, 3-outcome sport (Win/Draw/Loss) where a random guess is 33% and professional sportsbooks operate around 55%, this is a successful result.

Since the Elo difference feature is so dominant, the model struggles to predict underdog victories. When I tested the results to see if the model correctly predicted any of the actual underdog victories that happened in the 2022 World Cup (such as Morocco winning against Portugal), the result was zero.

Simulating the 2026 World Cup

For the final application of the models, I built a custom knockout bracket engine to simulate and predict the 2026 World Cup Champion. The 2026 tournament is expanding to 48 teams, meaning the eight best third-place teams will also advance from the groups. Simulating this correctly would require excessive custom tie-breaking code. Because of this, I decided to start my simulation directly at the Round of 32. I made my own predictions on which 32 teams would survive the group stage and placed them into the correct brackets.

Because 2026 is hosted in North America, the simulator function automatically treats matches as neutral venues unless the USA, Canada, or Mexico are playing. For every matchup, the function grabs each team's final Elo rating and their most recent 5-game momentum statistics from the dataset. It builds a fresh data row, passes it through the XGBoost models to get Expected Goals, and uses the Poisson simulator to determine the winner. In the knockout stage, there are no draws. If the model predicts a draw as the highest probability, the engine automatically gives the win to the team with the higher chance of an outright win.

Here are the final simulation results for my 2026 bracket:

Note: While the model failed to predict underdog victories in the 2022 World Cup, it did predict some underdog victories for the 2026 tournament. For example, it predicted Portugal reaching the final despite having a lower Elo ranking than heavyweights they would face, such as Brazil

and Argentina. Since the tournament hasn't happened yet, only time will tell if these predictions are accurate.

- **Figure 9: 2026 Round of 32 Prediction**

```
# RUN THE BRACKET
round_of_16    <- simulate_round(round_of_32, "ROUND OF 32")
✓ 0.0s

--- ROUND OF 32 ---
Germany vs Paraguay  ->  Winner: Germany
France vs Tunisia  ->  Winner: France
South Africa vs Canada  ->  Winner: Canada
Netherlands vs Morocco  ->  Winner: Netherlands
Colombia vs Croatia  ->  Winner: Colombia
Spain vs Algeria  ->  Winner: Spain
United States vs Norway  ->  Winner: United States
Belgium vs South Korea  ->  Winner: Belgium
Brazil vs Japan  ->  Winner: Brazil
Ivory Coast vs Senegal  ->  Winner: Senegal
Mexico vs Cape Verde  ->  Winner: Mexico
England vs DR Congo  ->  Winner: England
Argentina vs Uruguay  ->  Winner: Argentina
Turkey vs Egypt  ->  Winner: Turkey
Switzerland vs Austria  ->  Winner: Switzerland
Portugal vs Ghana  ->  Winner: Portugal
```

- **Figure 10: 2026 Round of 16 Prediction**

```
quarter_finals <- simulate_round(round_of_16, "ROUND OF 16")
✓ 0.0s

--- ROUND OF 16 ---
Germany vs France  ->  Winner: France
Canada vs Netherlands  ->  Winner: Canada
Colombia vs Spain  ->  Winner: Spain
United States vs Belgium  ->  Winner: Belgium
Brazil vs Senegal  ->  Winner: Brazil
Mexico vs England  ->  Winner: England
Argentina vs Turkey  ->  Winner: Argentina
Switzerland vs Portugal  ->  Winner: Portugal
```

- **Figure 11: 2026 Quarter-Finals Prediction**

```
semi_finals    <- simulate_round(quarter_finals, "QUARTER-FINALS")
✓ 0.0s

--- QUARTER-FINALS ---
France vs Canada  ->  Winner: France
Spain vs Belgium  ->  Winner: Spain
Brazil vs England  ->  Winner: Brazil
Argentina vs Portugal  ->  Winner: Portugal
```

- **Figure 12: 2026 Semi-Finals Prediction**

```
final_match    <- simulate_round(semi_finals, "SEMI-FINALS")
✓ 0.0s

--- SEMI-FINALS ---
France vs Spain  ->  Winner: Spain
Brazil vs Portugal  ->  Winner: Portugal
```

- **Figure 13: 2026 Finals Prediction**

```
champion       <- simulate_round(final_match, "WORLD CUP FINAL")
✓ 0.0s

--- WORLD CUP FINAL ---
Spain vs Portugal  ->  Winner: Spain
```

5. Conclusions and Discussion

Overall Conclusions

This project proved that a machine learning pipeline using XGBoost and a Poisson distribution effectively predicts international football outcomes. By predicting Expected Goals instead of a simple win/loss, the model successfully captured the goal differentials needed for World Cup simulations. However, historical team strength heavily dominated recent momentum. When tested against the 2022 World Cup, the model failed to predict any actual underdog victories, such as Morocco's historic run.

Interpreting the Models and Results

Feature importance plots show the strength gap dictates match flow more than any other variable. While the advanced model uses 5-game momentum to adjust probabilities, this momentum was never mathematically strong enough to flip predictions for heavy underdogs in 2022. However, for the 2026 simulation, the model predicted Portugal reaching the final despite facing higher-ranked teams like Brazil and Argentina. This proves it recognizes when lower-ranked teams have the right momentum to win.

What Changed During the Project

- **Data Filtering:** I restricted training data to matches from 1998 onward because early football history had wildly inconsistent scoring rates.
- **Model Choices:** I initially planned to only train an advanced model, but I added a Baseline Elo Model to mathematically prove my hypothesis that adding 5-game momentum reduces prediction error.

- **Simulation Scope:** I originally wanted to simulate the entire 2026 World Cup. Because the expanded 48-team format requires complex tie-breaking logic for third-place teams, I changed the scope to start directly at the Round of 32 using my own predictions.

Limitations and Future Work

The current model relies purely on macro-level team statistics and ignores micro-level features like star player injuries, squad rotations, or individual brilliance. In a subsequent exploration, I would improve the model by integrating player-level statistics (such as average squad age or total team market value) to capture on-pitch talent. Second, I would use a Bivariate Poisson distribution instead of two independent Poisson models to capture the correlation between home and away goals (this can help capture patterns such as early goals changing opponents strategies).

6. References

- **Dataset:** Jürisoo, M. (2026). *International football results from 1872 to the present*. Kaggle. https://www.kaggle.com/datasets/martj42/international-football-results-from-1872-to-2017
- **Technical Article:** Pena, L. (2021). *Predicting Football Results With the Poisson Distribution*. https://pena.lt/y/2021/06/18/predicting-football-results-using-the-poisson-distribution/
- **Paper:** Chen, T., & Guestrin, C. (2016). *XGBoost: A Scalable Tree Boosting System*.
- **R Packages:**, elo, ggplot2, corrplot, xgboost
- **Code Repository:** Hassan, Y. (2026). *World Cup 2026 Prediction Pipeline*. GitHub. https://github.com/Youssef-Chip/data-analytics/blob/main/final_project/final_project.ipynb